

HOUSTON HOOKS, TX, USA

WMO: 722429

Lat: 30.068N

Lon: 95.556W

Elev: 152

StdP: 14.62

Time Zone: -6.00 (NAC)

Period: 98-19

WBAN: 53910

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.8	33.4	16.6	12.8	41.9	20.8	15.7	44.6	20.0	59.4	18.4	59.2	3.7	0	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.5	98.1	75.9	95.8	76.4	93.5	76.5	80.2	88.3	79.4	87.9	78.8	87.3	6.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.7	149.8	82.5	77.4	143.3	82.2	76.8	140.4	82.0	43.6	87.7	43.0	87.5	42.4	87.5	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.5	15.8	13.4	24.2	101.9	4.0	4.1	21.3	104.9	19.0	107.3	16.8	109.6	13.9	112.6
			22.6	81.3	3.4	0.9	20.2	82.0	18.2	82.5	16.4	83.1	13.9	83.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	69.6	52.9	56.8	62.9	69.4	76.5	82.0	83.6	84.2	79.8	70.7	61.1	54.0
	DBStd	13.31	9.66	9.86	8.62	6.69	5.18	3.19	2.79	3.09	4.53	7.38	8.98	9.54
	HDD50	209	79	42	10	1	0	0	0	0	0	0	14	63
	HDD65	1404	391	260	144	34	2	0	0	0	0	33	177	363
	CDD50	7351	169	233	410	583	823	960	1042	1060	894	642	347	188
	CDD65	3071	16	31	79	167	360	510	577	595	444	209	60	23
	CDH74	31194	52	128	427	1174	3301	5622	6712	7007	4372	1906	397	96
	CDH80	13665	1	11	57	245	1173	2585	3289	3608	2003	628	60	5
Wind	WSAvg	5.9	6.7	7.0	7.1	7.1	6.6	5.5	4.9	4.6	4.6	5.1	5.6	6.2
Precipitation	PrecAvg	45.8	3.6	3.0	3.3	3.5	4.6	4.5	3.2	3.2	3.9	4.9	4.3	3.6
	PrecMax	63.8	10.0	7.1	8.1	11.5	11.5	15.9	11.3	9.1	7.4	17.0	12.6	9.5
	PrecMin	17.0	0.4	0.4	0.6	0.2	0.0	0.0	0.0	0.0	0.9	0.0	0.6	1.0
	PrecStd	11.4	2.2	1.9	1.9	2.5	3.3	3.5	2.8	2.4	1.9	4.6	2.7	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.0	81.5	84.7	88.3	93.3	98.7	99.6	101.6	99.9	91.1	84.8	80.5
		MCWB	66.0	66.8	68.4	71.6	74.4	76.0	75.5	75.2	73.9	73.8	72.7	69.7
	2%	DB	75.0	78.0	81.4	85.0	90.6	95.1	97.1	98.5	94.9	89.1	81.5	76.7
		MCWB	65.3	66.5	68.4	70.6	74.3	75.9	76.3	76.4	74.7	74.3	70.0	68.6
	5%	DB	72.3	74.7	79.1	82.5	88.5	92.8	94.7	96.0	92.3	86.5	78.9	73.5
		MCWB	64.7	65.8	67.0	69.7	74.3	76.2	76.8	77.0	75.2	72.9	68.4	67.5
	10%	DB	69.7	72.0	75.7	80.9	86.5	90.8	92.6	93.4	90.1	83.6	75.4	71.0
		MCWB	63.8	65.0	65.7	68.7	73.7	76.3	77.1	77.1	74.9	71.1	67.9	65.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.3	71.1	72.4	75.8	78.4	80.5	80.7	81.1	80.3	78.8	75.2	72.8
		MCDB	73.5	75.8	79.4	82.9	87.2	89.3	88.5	88.6	86.7	84.7	80.2	76.4
	2%	WB	68.5	69.6	71.0	74.0	77.2	79.4	80.0	80.1	79.1	77.4	73.4	71.0
		MCDB	72.2	74.6	77.2	80.5	86.3	87.5	88.3	88.5	86.2	83.7	77.9	74.7
	5%	WB	67.0	68.3	69.9	72.9	76.2	78.5	79.2	79.4	78.2	76.2	71.7	69.4
		MCDB	71.0	72.8	75.4	79.1	84.5	86.7	88.0	88.4	85.8	82.4	76.4	73.2
	10%	WB	64.7	66.7	68.7	71.6	75.3	77.9	78.5	78.6	77.4	74.8	69.9	66.7
		MCDB	68.5	70.8	74.0	77.5	83.0	86.1	87.1	87.8	85.2	80.6	74.3	70.2
Mean Daily Temperature Range	5% DB	MDBR	20.0	19.5	20.0	19.4	18.3	17.6	18.2	19.5	19.6	21.8	21.1	20.0
		MCDBR	21.8	20.4	19.9	19.6	19.2	20.4	21.5	22.5	22.7	22.2	22.1	22.0
		MCWBR	13.4	11.5	9.4	8.5	6.9	5.5	5.4	5.5	7.0	9.1	11.8	14.8
	5% WB	MCDDBR	17.8	17.0	16.9	15.4	16.4	16.4	17.8	19.0	17.7	17.8	18.3	19.1
		MCWBR	13.1	11.7	10.2	8.3	6.8	5.7	5.9	5.8	6.4	9.0	12.1	15.2
Clear-Sky Solar Irradiance	taub	0.337	0.358	0.374	0.433	0.481	0.468	0.493	0.480	0.450	0.380	0.372	0.343	
	taud	2.498	2.451	2.401	2.237	2.140	2.231	2.182	2.221	2.267	2.451	2.416	2.488	
	Ebn at Noon	284	287	289	274	260	262	255	258	262	276	268	276	
	Edh at Noon	29	33	36	45	49	45	47	45	41	32	31	28	
All-Sky Solar Radiation	RadAvg	909	1021	1395	1681	1853	1947	1938	1853	1590	1359	1020	800	
	RadStd	88	185	123	115	138	159	167	141	158	170	105	98	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+311	+239
Regional (37 neighbors)		+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+196	+171

Nomenclature: See separate page